

Square and Phase Norm Equalities on $C_0(L)$

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Status. Candidate strong partial result, likely valid, pending human review. The packet completely classifies both real square identities and every complex fixed-phase identity from the source on the classical spaces $C_0(L)$. The source's arbitrary-Banach-space questions remain open.

1 Source questions

Kadets, Martín, and Merí study norm identities required of every rank-one operator on a Banach space [1]. In the real case they say that the nonsurjective analytic-function case is unclear and single out $g(t) = t^2$ and $g(t) = -t^2$. The resulting identities are

$$\|\text{Id} + T^2\| = 1 + \|T^2\|, \quad \|\text{Id} - T^2\| = 1 + \|T^2\|. \quad (1)$$

Their source Proposition 4.8 characterizes these identities geometrically, but does not identify the spaces satisfying them. In the complex case they study, for $\omega \in \mathbb{T} \setminus \{1\}$,

$$\|\text{Id} + \omega T\| = \|\text{Id} + T\| \quad \text{for every rank-one } T, \quad (2)$$

and write that they do not know whether the infinitely many subgroup properties are all equivalent.

The theorem below gives a complete answer to all these questions on real and complex $C_0(L)$ spaces, equivalently on commutative C^* -algebras and AM-spaces represented in this form.

2 Classification theorem

Let L be a locally compact Hausdorff space and let $C_0(L, \mathbb{K})$ carry the supremum norm. We exclude the one-point real space in the first statement: there T^2 is a nonnegative scalar, so the positive-square identity is automatic although the Daugavet property fails.

Theorem 1 (Real square classification). *Suppose $|L| \geq 2$ and put $X = C_0(L, \mathbb{R})$. The following are equivalent.*

1. $\|\text{Id} + T^2\| = 1 + \|T^2\|$ for every rank-one $T \in \mathcal{L}(X)$.
2. $\|\text{Id} - T^2\| = 1 + \|T^2\|$ for every rank-one $T \in \mathcal{L}(X)$.
3. L has no isolated points.
4. X has the Daugavet property.

On the other hand, we do not know if a result similar to the above theorem is true when the function g is not onto. Let us give some remarks about two easy cases:

$$g(t) = t^2 \quad (t \in \mathbb{R}) \quad \text{and} \quad g(t) = -t^2 \quad (t \in \mathbb{R}).$$

In the first case, it is easy to see that if the norm equality

$$\|\text{Id} + T^2\| = f(\|T^2\|)$$

holds for every rank-one operator, then $f(t) = 1 + t$ and, therefore, the interesting norm equality in this case is

$$(19) \quad \|\text{Id} + T^2\| = 1 + \|T^2\|.$$

This equation is satisfied by every rank-one operator T on a Banach space X with the Daugavet property. Let us also recall that the equality

$$|1 + t^2| = 1 + |t^2|$$

holds for every $t \in L(\mathbb{R}) \equiv \mathbb{R}$ (Remark 4.2). For the norm equality

$$\|\text{Id} - T^2\| = f(\|T^2\|),$$

we are not able to get any information about the shape of the function f . Going to the 1-dimensional case, we get that

$$|1 - t^2| = \max\{1 - |t^2|, |t^2| - 1\}$$

Figure 1: The real nonsurjective case and the two square identities in the source PDF, pages 16–17.

Theorem 2 (Complex phase classification). *Let $X = C_0(L, \mathbb{C})$ and fix any $\omega \in \mathbb{T} \setminus \{1\}$. Then (2) holds for every rank-one $T \in \mathcal{L}(X)$ if and only if L has no isolated points, equivalently if and only if X has the Daugavet property. Consequently every finite-root property and the full-circle property in the source are equivalent on complex $C_0(L)$ spaces.*

3 The no-isolated-points direction

We include the short local-bump proof so that the classification is self-contained.

Lemma 1. *For $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, if L has no isolated points, then $C_0(L, \mathbb{K})$ has the Daugavet property.*

Proof. We verify the standard slice characterization. Let $f \in S_{C_0(L)}$, $\mu \in S_{C_0(L)^*}$, and $\varepsilon > 0$. Choose a unit scalar $\zeta \in \mathbb{K}$ and a nonempty open set U such that

$$\text{Re}(\bar{\zeta}f(t)) > 1 - \varepsilon \quad (t \in U).$$

Every nonempty open subset of a locally compact Hausdorff space without isolated points is uncountable: otherwise that open Baire space would be a countable union of nowhere dense singletons. A finite regular measure has at most countably many atoms, so choose a non-atom $t \in U$ of $|\mu|$. Regularity gives an open neighborhood $V \subset U$ of t with $|\mu|(V) < \varepsilon$.

5. THE NEW PROPERTIES

Our aim in this section is to give some remarks concerning the properties appearing at Theorems 4.5 and 4.7b, i.e. we consider a non-trivial multiplicative subgroup A of $\mathbb{T}$ and study those Banach spaces for which all rank-one operators satisfy the norm equality

$$\|\text{Id} + \omega T\| = \|\text{Id} + T\|.$$

In the real case, only one property arises; in the complex case, there are infinitely many properties and we do not know if all of them are equivalent.

Our first (easy) observation is that all these properties pass from the dual of a Banach space to the space. We will see later that the converse result is

Figure 2: The complex equivalence question in the source PDF, page 18.

Choose $y_0 \in B_{C_0(L)}$ with $\text{Re } \mu(y_0) > 1 - \varepsilon$, and choose $\varphi \in C_c(L)$ such that $0 \leq \varphi \leq 1$, $\varphi(t) = 1$, and $\text{supp } \varphi \subset V$. Put

$$y = (1 - \varphi)y_0 + \varphi\zeta.$$

Pointwise convexity of the scalar unit ball gives $\|y\| \leq 1$, while

$$|\mu(y - y_0)| \leq 2|\mu|(V) < 2\varepsilon.$$

Thus y remains in an arbitrarily thin slice determined by μ , and

$$\|f + y\| \geq |f(t) + \zeta| \geq 1 + \text{Re}(\bar{\zeta}f(t)) > 2 - \varepsilon.$$

After harmless rescaling of the three occurrences of ε , this is the slice characterization of the Daugavet property. $\square$

If L has no isolated points, Lemma 1 applied to the rank-one operators T^2 , $-T^2$, T , and ωT proves all forward implications in Theorems 1 and 2.

4 Exact isolated-point obstructions

Suppose now that $p \in L$ is isolated and choose $q \in L \setminus \{p\}$. The function $e = \mathbf{1}_{\{p\}}$ lies in $C_0(L)$. Complete regularity gives $g \in C_0(L, \mathbb{R})$ with

$$\|g\| = 1, \quad g(p) = 0, \quad g(q) = 1.$$

Set

$$f = e + \frac{1}{2}g, \quad \phi = -\frac{1}{4}\delta_p + \frac{3}{4}\delta_q, \quad T = \phi \otimes f, \tag{3}$$

where $(\phi \otimes f)(h) = \phi(h)f$. Then

$$\|f\| = \|\phi\| = \|T\| = 1, \quad \phi(f) = \frac{1}{8}, \quad T^2 = \frac{1}{8}T.$$

Evaluations norm $C_0(L)$, so

$$\|\text{Id} + T^2\| = \sup_{t \in L} \left\| \delta_t + \frac{1}{8}f(t)\phi \right\|.$$

At p and q the two total-variation norms are respectively

$$|1 - \frac{1}{32}| + \frac{3}{32} = \frac{17}{16}, \quad \frac{1}{64} + |1 + \frac{3}{64}| = \frac{17}{16}.$$

For $t \notin \{p, q\}$, the three point masses are disjoint and $|f(t)| \leq 1/2$, giving the upper bound

$$1 + \frac{1}{8}|f(t)|\|\phi\| \leq \frac{17}{16}.$$

Therefore

$$\boxed{\|\text{Id} + T^2\| = \frac{17}{16} < \frac{18}{16} = 1 + \|T^2\|.} \quad (4)$$

This disproves the positive-square identity whenever an isolated point is present.

For the negative-square identity, use the rank-one projection $P = \delta_p \otimes e$. Since $P^2 = P$ and q exists,

$$\|\text{Id} - P^2\| = 1 < 2 = 1 + \|P^2\|.$$

This completes Theorem 1.

Over $\mathbb{C}$, the same projection satisfies

$$\|\text{Id} + P\| = 2, \quad \|\text{Id} + \omega P\| = \max\{1, |1 + \omega|\} < 2 \quad (\omega \neq 1),$$

which completes Theorem 2.

5 Scope and novelty audit

The result is full for all real and complex $C_0(L)$ spaces and gives a single topological criterion for every source identity considered here. It does not prove that the positive-square identity forces the Daugavet property on an arbitrary real Banach space, nor does it settle equivalence of the complex subgroup properties on arbitrary spaces.

Bounded searches covered the run indexes, the local full-source corpus, official arXiv, and web results. Queries used the source id and title, the two square identities, “rank-one square Daugavet”, “extremely non-complex”, and the fixed-phase identity. Later papers [2, 3] construct extremely non-complex spaces and cite the source, but do not state the classification above; their function-space examples use perfect ambient compacta and lie on the positive side of the same local-bump argument. No primary source stating Theorems 1–2 was located. This is bounded evidence, not a novelty certificate.

The script `code/check_obstruction.py` checks the exact rational arithmetic in (4). Human review should focus on the slice argument in Lemma 1, the $C_0(L)^*$ total-variation computation, and the singleton exception.

References

- [1] V. Kadets, M. Martín, and J. Merí, *Norm equalities for operators*, arXiv:math/0604102 (2006); Indiana Univ. Math. J. 56 (2007), 2385–2411.
- [2] P. Koszmider, M. Martín, and J. Merí, *Extremely non-complex $C(K)$ spaces*, arXiv:0811.0577 (2008); J. Math. Anal. Appl. 350 (2009), 601–615.
- [3] P. Koszmider, M. Martín, and J. Merí, *Isometries on extremely non-complex Banach spaces*, arXiv:0901.1512 (2009); J. Inst. Math. Jussieu 10 (2011), 325–348.